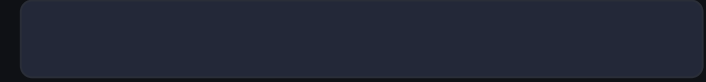




AI SRE Copilot



MULTI-AGENT AI SRE PLATFORM

Autonomous incident investigation for Kubernetes

Alerts fire, the platform investigates on its own, produces a grounded root-cause report and an execution-ready runbook — and a human approves anything that touches the cluster.



SRE toil doesn't scale with headcount

01

Alerts outpace attention

A cascading outage can fire dozens of alerts in minutes. Someone still has to open Grafana, Jaeger, and kubectl, by hand, for every one.

02

Root cause lives across three tools

Metrics say what broke. Traces say where. Kubernetes events say why. No single dashboard correlates all three under time pressure.

03

Tribal knowledge doesn't transfer

The engineer who remembers last quarter's near-identical incident isn't always the one on call this time.

What the platform actually does



Autonomous detection

Alertmanager posts a firing alert to the webhook; the copilot investigates with nobody watching, no polling.



Parallel multi-agent investigation

Metrics, Trace, and Kubernetes agents gather evidence at once — every tool call is visible, not hidden.



Grounded root-cause reports

The RCA is tied to the actual metrics and traces pulled for this incident, not a generic response.



Execution-ready runbooks

A second structured LLM pass turns the RCA into concrete, ready-to-run remediation steps.



Human-gated remediation

The one cluster-mutating action pauses on a real approval gate — nothing touches the cluster without a person deciding.



Incident memory

Semantic search over past incidents answers “has this happened before?” instead of starting cold.

What we'll walk through

1 Trigger

kubectl scale deployment product-catalog --replicas=0 — a real failure in the live cluster.

2 Auto-triage

Alertmanager posts to the webhook; the copilot picks it up with nobody asking, no polling.

3 Investigate

Metrics, trace, and Kubernetes agents gather evidence in parallel; the coordinator correlates it.

4 Root cause + runbook

An LLM produces a grounded RCA and an execution-ready remediation runbook.

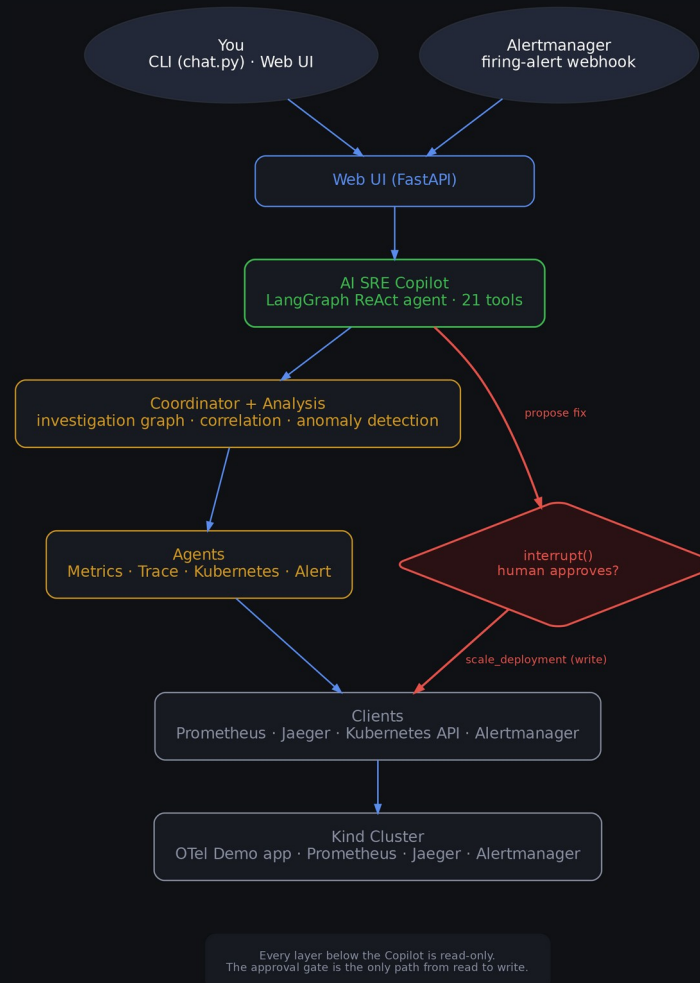
5 Approve

The proposed fix parks on awaiting_approval until a human clicks Approve — in the Web UI or CLI.

6 Resolve

Operations Hub's KPIs and the Timeline update live — resolution rate, time-to-mitigate, the works.

Six layers, one approval gate



Beyond the core loop



Alert correlation

Groups related alerts into one incident instead of a dozen separate pings for the same outage.



Anomaly detection

Z-score checks flag a metric behaving unusually against that service's own history.



Knowledge base search

Semantic / TF-IDF search over human-authored runbooks and playbooks, not just past incidents.



Operations Hub

KPI dashboard — resolution rate, time-to-mitigate, pending approvals — at a glance.



Google sign-in + PII redaction

Real per-user auth, plus a regex scrub before cluster data reaches the LLM or gets stored.



CLI copilot

Same agent, same tools, from a terminal instead of the browser.

Stated plainly: what's next

ACCEPTED TRADE-OFFS

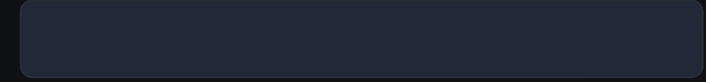
- No per-role authorization — every allowlisted user can also approve a remediation.
- PII redaction is pattern-based, not a compliance-grade classifier.
- OAuth client secret and session key still live in plaintext .env.
- Token refresh isn't implemented — sessions simply expire after 8 hours.

PLANNED NEXT

- Per-role authorization — separate “can view” from “can approve a remediation.”
- Slack / Microsoft Teams notifications.
- MCP server support.
- True embedding-based search everywhere — Voyage AI is optional today, TF-IDF is the fallback.



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Questions

AI SRE Copilot — Multi-Agent AI SRE Platform — thank you.